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Non-invasive detection of coronary artery disease in high-risk patients based on the stenosis prediction of separate coronary arteries

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ABSTRACT

Background and objective: Cardiovascular diseases are an extremely widespread sickness and account for 17 million deaths in the world per annum. Coronary artery disease (CAD) is one of such diseases with an annual mortality rate of about 7 million. Thus, early diagnosis of CAD is of vital importance. Angiography is currently the modality of choice for the detection of CAD. However, its complications and costs have prompted researchers to seek alternative methods via machine learning algorithms.
Methods: The present study proposes a novel machine learning algorithm. The proposed algorithm uses three classifiers for detection of the stenosis of three coronary arteries, i.e., left anterior descending (LAD), left circumflex (LCX) and right coronary artery (RCA) to get higher accuracy for CAD diagnosis.
Results: This method was applied on the extension of Z-Alizadeh Sani dataset which contains demographic, examination, ECG, and laboratory and echo data of 500 patients. This method achieves an accuracy, sensitivity and specificity rates of 96.40%, 100% and 88.1%, respectively for the detection of CAD. To our knowledge, such high rates of accuracy and sensitivity have not been attained elsewhere before.
Conclusion: This new algorithm reliably distinguishes those with normal coronary arteries from those with CAD which may obviate the need for angiography in the normal group.
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1. Introduction

CORONARY artery disease (CAD) is deemed a major type of cardiovascular diseases and is responsible for high annual mortality rates around the globe. The widespread and particular potency of CAD, necessitate early diagnosis [1].

To that end, various diagnostic modalities have been developed over the years, which are divided into two categories of invasive and non-invasive [2]. The first category, which includes angiography, inherently carries high risk and the second one, which comprises electrocardiography (ECG) and echocardiography, lacks desirable accuracy and reliability. Angiography is currently regarded as the optimal diagnostic tool for CAD. Nevertheless, its invasive nature may be life-threatening for patients.

Indeed, angiography has been reported to be the culprit for 0.1% to 0.14% mortality, 0.06% to 0.07% myocardial infarction, 0.07% to 0.14% brain stroke, 0.23% reaction to contrast material, and 2% local vascular problems [3]. Given the aforementioned complications allied to angiography, research has been undertaken to reduce the need for angiography in individuals suspected for CAD.

Recently, machine learning and data mining algorithms are widely used in disease diagnosis such as breast cancer [4,5], liver disease [6], coronary calcium lesion [7–9] and some other diseases [10–31]. Data mining algorithms, which seek hidden and gainful relationships in datasets, have proved to be extremely useful tools in the pursuit of this objective [32].

Needless to say, data mining requires collection of information, which in the setting of CAD, encompasses the data of exercise test [33,34], clinical information [35–37], Doppler signals [38–43], heart sound [44, 45], ECG signal [46–49], and Features of Heart Rate Variability [50,51].

Most of the works conducted thus far have drawn upon the UCI Heart disease datasets [52–59]. However, they are not up to date. Their last modifications were made before 1992. In other researches, the datasets are not contain some important features

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such as ECG, laboratory and Echo together. Most of them had fewer samples in their datasets. The more important drawback of previous works is that they have not considered the impact of LAD, LCX and RCA stenosis for diagnosis of CAD. Meanwhile, as they did not use some important features, their performance is not satisfactory.

We seek to present a new, low-cost, uncomplicated, and accurate algorithm for early diagnosis of CAD based on data mining methods on extended Z-Alizadeh Sani dataset. Z-Alizadeh Sani dataset has been collected in our previous works [60]. As far as we know, for the first time in the literature, we used LAD, LCX and RCA stenosis prediction for CAD diagnosis. Meanwhile, we used an up-to-date dataset having 500 samples. We also achieved to the highest accuracy and sensitivity rates for CAD detection. Results of angiography were used as truth in supervised machine learning algorithm to predict stenosis based on non-invasively acquired data.

2. Medical dataset used

We used extended version of Z-Alizadeh Sani dataset containing the information of 500 patients. In the previous version, there were 303 record of patients [60]. However, in the new one, it is extended to 500 records to achieve more confident results. In this extended dataset, 351 patients have CAD and 149 patients are non-CAD. It means that 149 non-CAD subjects were combination of healthy subjects and non-CAD disease patients. “non-CAD disease patients” means that they may have some kind of diseases but these diseases are not CAD. Meanwhile, we collected features about people who refereed to clinic and then we used angiography to determine CAD and non-CAD patients. In our dataset, target label was distinguished by angiography test result. If stenosis in at least one of LAD, LCX or RCA arteries is more than 50%, the patient has CAD.

The dataset contains 54 features with no missing value listed in Table 1. The features are divided into four different categories: Demographic, symptom and examination, ECG, laboratory and echo. For each patient, the dataset has three more labels in addition to CAD label. It contains whether each of the coronary arteries LAD, LCX and RCA are stenotic or not. In our proposed method, we use this additional information to enhance the classifier’s performance. Left main coronary artery (LMCA) not been considered in this study for CAD detection because the probability of stenosis of this artery is lower than other coronary arteries (LAD, LCX and RCA) [61]. Meanwhile, some of patients who referred to clinic having stenosis of left main coronary artery were in emergency condition and were sent for angiography very soon. So, we did not have enough time to get other tests and we could not collect 54 features.

3. Method

For accurate CAD diagnosis, it is sufficient to know the stenosis of the LAD, LCX and RCA arteries. Data mining diagnosis methods can be applied to predict the stenosis of the individual arteries. Indeed, we use three classifiers for stenosis prediction of each coronary artery. These classifiers are called C_{LAD}, C_{LCX}, and C_{RCA}.

In the rest of this section, we first discuss our base methods: single-classifier and three-classifier methods. Then, we introduce our proposed feature engineering method. In all of these methods, we have done feature selection based on weight-by-SVM [62]. In Weight by SVM, coefficients of the normal vector of a linear SVM are used as attribute weights. This operator works not only for two classes but also for multiple classes. Attribute values must be numerical.

Meanwhile, to calculate the accuracy, we used 10-fold cross validation in all of these three methods.

3.1. Single-classifier method

Single-classifier method is our first base method. It uses all the features directly for CAD diagnosis. In this method, the stenosis of individual arteries is not considered. A single classifier that is named C_S is trained on the features. The classifier is used for CAD diagnosis of new patients. For a patient p, C_S is 1, if the classifier predicted that p has CAD. Otherwise, it equals to 0.

3.2. Three-classifier method

Three-classifier method is our second base method. In this method, classification is only done according to diagnosis of LAD, LCX, and RCA stenosis because having CAD depends on the stenosis of the individual arteries.

We first train three classifiers named C_{LAD}, C_{LCX} and C_{RCA} to predict LAD, LCX and RCA stenosis respectively. The output of C_{LAD}, C_{LCX} and C_{RCA} classifiers are binary variables showing the stenosis of the three coronary arteries base on the features listed in Table 2. To select the best features for each classifier, we ranked the features by Weight-by-SVM feature selection method. Then, we selected the first K features. To avoid over fitting for selecting the best value for K, we are not allowed to see test data. So, we select the best value for K only based on training data. To do so, we divided the training data to training (70%) and validation (30%) data. After selection the best value for K base on best performance on validation data, we train again on the whole training and validation data with the selected K.

Then, the final classifier C_T only uses the output of these three classifiers to predict CAD. Classifier C_T is defined as Eq. (1).

$$C_T(p) = C_{LAD}(p) \vee C_{LCX}(p) \vee C_{RCA}(p) \tag{1}$$

where $C_T(p)$ is a binary variable which determines having CAD for patient p. Based on Eq. (1), C_T predicts that a patient has CAD, if any of the coronary arteries are predicted as stenotic.

The error probability of C_T is shown in Eq. (2).

$$\begin{aligned} error(C_T) &= FP(C_T) \times Pr(CAD = 0) + FN(C_T) \times Pr(CAD = 1) \\ &= \Pr(X \cup Y \cup Z | CAD = 0) \times \Pr(CAD = 0) \\ &\quad + \Pr(X \cap Y \cap Z | CAD = 1) \times \Pr(CAD = 1) \end{aligned} \tag{2}$$

where X, Y and Z are stochastic events showing the wrong prediction of stenosis of LAD, LCX and RCA arteries, respectively.

The term $\Pr(X \cup Y \cup Z \mid CAD = 0)$ equals to the probability that a healthy patient be predicted to have at least one stenotic artery and the term $\Pr(X \cap Y \cap Z | CAD = 1)$ equals to the probability that all three classifiers make mistake in predicting the stenosis of their corresponding arteries. The probabilities of Eq. (2) can be estimated by comparing the prediction of the C_{LAD}, C_{LCX}, and C_{RCA} classifiers and the truth values. However, as Eq. (1) uses “OR” operator, it is expected that $\Pr(X \cup Y \cup Z \mid CAD = 0)$ (false positive) be relatively high and the $\Pr(X \cap Y \cap Z | CAD = 1)$ (false negative) be relatively low. Therefore, the false positive and false negative estimations in Eq. (2) will be biased. We will propose our feature engineering method in the next subsection to resolve the above issue.

3.3. Feature engineering method (The proposed method)

In this section, we introduce our proposed feature engineering method. Although knowing the stenosis of the arteries can determine whether the patient has CAD or not, using only three-classifier method may not lead to the desired accuracy. The reason is that naively combining the results of the C_{LAD}, C_{LCX}, and C_{RCA} classifiers as in Eq. (1) will lead to a classifier with biased false positive and false negatives. In contrast, the feature engineering method uses the results of the three classifiers to improve the accuracy, while it will not have the above issue.

Table 1
Features of Z-Alizadeh Sani dataset.

Feature type	Feature name	Range
Demographic	Age	30–86
	Weight	48–120
	Sex	Male, Female
	BMI (Body Mass Index Kg/m^2)	18–41
	DM (Diabetes Mellitus)	Yes, No
	HTN (Hyper Tension)	Yes, No
	Current Smoker	Yes, No
	Ex-Smoker	Yes, No
	FH (Family History)	Yes, No
	Obesity	Yes if MBI > 25, No otherwise
	CRF (Chronic Renal Failure)	Yes, No
	CVA (Cerebrovascular Accident)	Yes, No
	Airway Disease	Yes, No
	Thyroid Disease	Yes, No
	CHF (Congestive Heart Failure)	Yes, No
	DLP (Dyslipidemia)	Yes, No
Symptom and examination	BP (Blood Pressure: mmHg)	90–190
	PR (Pulse Rate) (ppm)	50–110
	Edema	Yes, No
	Weak peripheral pulse	Yes, No
	Lung Rales	Yes, No
	Systolic murmur	Yes, No
	Diastolic murmur	Yes, No
	Typical Chest Pain	Yes, No
	Dyspnea	Yes, No
	Function Class	1, 2, 3, 4
	Atypical	Yes, No
	Nonanginal CP	Yes, No
	Exertional CP (Exertional Chest Pain)	Yes, No
	Low Th Ang (low Threshold angina)	Yes, No
ECG	Rhythm	Sin, AF
	Q Wave	Yes, No
	ST Elevation	Yes, No
	ST Depression	Yes, No
	T inversion	Yes, No
	LVH (Left Ventricular Hypertrophy)	Yes, No
	Poor R Progression (Poor R Wave Progression)	Yes, No
Laboratory and echo	FBS (Fasting Blood Sugar) (mg/dl)	62–400
	Cr (creatine) (mg/dl)	0.5–2.2
	TG (Triglyceride) (mg/dl	37–1050
	LDL (Low density lipoprotein) (mg/dl)	18–232
	HDL (High density lipoprotein) (mg/dl)	15–111
	BUN (Blood Urea Nitrogen) (mg/dl)	6–52
	ESR (Erythrocyte Sedimentation rate) (mm/h)	1–90
	HB (Hemoglobin) (g/dl)	8.9–17.6
	K (Potassium) (mEq/lit)	3.0–6.6
	Na (Sodium) (mEq/lit)	128–156
	WBC (White Blood Cell) (cells/ml)	3700–18,000
	Lymph (Lymphocyte) (%)	7–60
	Neut (Neutrophil) (%)	32–89
	PLT (Platelet) (1000/ml)	25–742
	EF (Ejection Fraction) (%)	15–60
	Region with RWMA[a] (Regional Wall Motion Abnormality)	0,1,2,3,4
	VHD (Valvular Heart Disease)	Normal, Mild, Moderate, Severe

[a] 17-segmet model was recommended by ASE be used to assign the units of analysis of wall motion. Walls were scored as 1: normal, 2: hypokinetic, 3: akinetic, 4: dyskinetic, and 5: aneurysmal [63].

In this method, we first train three classifiers on selected features for the three arteries. For feature selection, different methods such as information gain [64], filter method [57], wrapper method [65] and weight-by-SVM were tested. Weight-by-SVM had the best performance among these methods. For LAD, LCX and RCA diagnosis stenosis, 24, 22 and 25 features were selected respectively. The selection of these features was similar to feature selection algorithm used in three-classifier method. These features are listed in Table 2. We used the results of these three classifiers as three new features.

In addition to these classifiers, a fourth classifier is created on the training dataset for diagnosing CAD. We use the posterior probability of having CAD as our fourth feature. Using the probabilities instead of the binary predictions for these four classifiers yields better discrimination of patients. These four new features are added to the training dataset.

Our final classifier named C_P determines CAD using both 28 important selected features (features 5 to 32 in Final Classifier column of Table 2) in the train step and the four added features. The output of C_p is the probability of having CAD. According to this probability, it is necessary to decide if a patient has or has not CAD. Therefore, we have to select a threshold to convert the output of the final classifier into binary. To select the best value for threshold, we are not allowed to see test data. So, we select the best value for threshold only based on training data. To do so, we divided the training data to training (70%) and validation (30%) data. After selection the best value for threshold base on validation

Table 2
Selected features for each classifier ordered by weight by SVM.

	C_{LAD} (LAD Classifier)	C_{LCX} (LCX Classifier)	C_{RCA} (RCA Classifier)	Final Classifier
1	Typical C. P.	Age	DM	Result of CAD Classifier
2	Atypical	Na	Age	Result of LAD Classifier
3	Age	Typical C. P.	ESR	Result of LCX Classifier
4	Region RWMA	CR	Poor R Progression	Result of RCA Classifier
5	Nonanginal CP	Function Class	Function Class	Typical C. P.
6	Diastolic Murmur	Region RWMA	Neut	Atypical
7	EF	BP	Region RWMA	Age
8	Tinversion	TG	Dyspnea	Nonanginal CP
9	PR	DM	Q Wave	DM
10	CVA	Sex	Atypical	Tinversion
11	HTN	EF	Sex	FH
12	FBS	Airway disease	Lymph	Region RWMA
13	Lymph	CRF	Typical C. P.	HTN
14	St Elevation	WBC	Nonanginal CP	TG
15	FH	Current Smoker	TG	PR
16	DM	EX-Smoker	BUN	Diastolic Murmur
17	Current Smoker	LVH	HTN	Current Smoker
18	Low TH Ang	Systolic Murmur	CR	Dyspnea
19	Function Class	Thyroid Disease	CRF	ESR
20	DLP	BUN	Diastolic Murmur	BP
21	Lung rales	K	HDL	Function Class
22	Dyspnea	FBS	LVH	Sex
23	St Depression		FH	FBS
24	EX-Smoker		VHD	St Depression
25			St Depression	St Elevation
26				Q Wave
27				DLP
28				K
29				Obesity
30				LVH
31				Low TH Ang
32				EF

data, we train again on the whole training and validation data with the selected threshold.

To classify test data, the first four classifiers are evaluated on them and four features are added to the set. Then, employing the 28 important and four added features, classifier C_P determines CAD in the test dataset.

The classifiers of the feature engineering method can be any type of classifiers with the property that they can predict the posterior probabilities in addition to the binary prediction of the labels. We have used SVM [62] in our experiments. The algorithm's flowchart is shown in Fig. 1.

4. Results

4.1. Evaluation of the proposed method

To evaluate the feature engineering method, 10-fold cross validation was used to divide the dataset into training and test sets and SVM classifier was used as the base classifier of the different methods. SVM was selected because this classifier showed the best performance on training dataset. We tested Linear, Sigmoid, RBF, and Polynomial as SVM kernels. Among these kernels, linear had the best performance on our dataset. Therefore, we used it as our SVM kernel.

P(CAD = 1) was estimated to be 71.28%. This probability is for visitors with chest pain and suspicious of having CAD who volunteered for angiography. After angiography, 71.28% of the visitors had CAD. The false positive and false negative probabilities of the single-classifier method were 9.97% and 26.85%, respectively. False positive and false negative probabilities of the three-classifier method were 18.52% and 20.13%, respectively. Error rates for each of LAD, LCX, and RCA classifiers were 21.60%, 25.09%, and 23.13%, respectively.

Finally, the false positive and false negative probabilities of the feature engineering method were calculated. This method led to a lower rate of false positive and false negative probabilities of 12.1% and 0%. In other words, we reached the sensitivity value of 100% and accuracy rate of 96.40%. We implemented the method using Java and Weka 3–4. Weka data mining algorithms, such as Naïve Bayes, C4.5, and SVM were used with their default parameters.

4.2. Testing different types of classifiers

To show our proposed method does not depend on specific classification algorithm, we used some famous classifiers such as SVM, C4.5 and Naive Bayes on it.

Table 3 shows the accuracy, sensitivity and specificity of the different classifiers on extended Z-Alizadeh Sani dataset. It shows the results for the single-classifier, three-classifier and our proposed feature engineering methods.

Table 3 also shows that our proposed feature engineering method increased the performance of traditional classifiers (Single-Classifier), significantly (about 10%). Meanwhile, it shows that the accuracy of C4.5 is higher than Naive Bayes but less than SVM.

Table 4 shows the accuracy rate of CAD diagnosis for different feature selection methods for different k values. In this table, for each feature selection method, we highlighted the most accuracy rate for different K values. Table 5 shows P-value of Wilcoxon rank-sum test [66] for accuracy of different feature selection methods and different K's with respect to the highest value. Weight-by-SVM with K = 28 had the highest accuracy rate. Area under receiver operating curve (ROC) of our proposed method has been shown in Fig. 2.

Table 3
Performance of different methods using different classifiers on the extended Z-Alizadeh Sani dataset.

	Algorithm	Accuracy (%)	Sensitivity (%)	Specificity (%)	AUC
Single-classifier	Naive Bayes	76.0 ± 4.4	80.0 ± 3.5	67.1 ± 5.5	0.80
	C4.5	80.0 ± 3.7	85.3 ± 3.6	67.4 ± 4.0	0.78
	SVM	85.0 ± 3.4	89.4 ± 4.1	76.8 ± 3.1	0.81
Three-classifiers	Naive Bayes	70.0 ± 4.6	76.8 ± 4.5	54.3 ± 5.4	0.65
	C4.5	78.8 ± 3.8	80.8 ± 2.9	73.1 ± 4.7	0.71
	SVM	81.0 ± 3.0	81.4 ± 2.7	79.8 ± 4.4	0.87
Feature engineering	Naive Bayes	86.0 ± 3.3	90.6 ± 2.6	73.6 ± 3.9	0.85
	C4.5	89.8 ± 3.5	91.1 ± 2.1	85.4 ± 3.8	0.83
	SVM	96.4 ± 2.7	100 ± 0	88.1 ± 2.8	0.92

Table 4
Accuracy rate of cad diagnosis for different feature selection methods for different k values.

	Feature selection algorithm			
K values	Weight by SVM	Information gain	Filter method	Wrapper method
5	85	85.7	80.1	84.1
10	86.9	88.1	82.3	85.2
15	88.1	89.3	83.7	85.2
20	91.3	90.5	87.6	86.8
25	93.4	89.8	91.6	87.1
28	96.4	88.7	90.8	89.9
30	95.1	87.4	90.8	93.3
35	94.6	87.3	90.5	93.3
40	94.6	86.4	89.3	92.9

Table 5
P-value of Wilcoxon rank-sum test with respect to the highest value according to Table 4.

	Feature selection algorithm			
K values	weight by SVM	information gain	filter method	wrapper method
5	0.00	0.00	0.00	0.00
10	0.00	0.00	0.00	0.00
15	0.00	0.00	0.00	0.00
20	0.01	0.00	0.04	0.00
25	0.34	0.00	0.04	0.00
28	–	0.00	0.00	0.00
30	0.58	0.00	0.00	0.04
35	0.44	0.00	0.00	0.02
40	0.44	0.00	0.00	0.02

4.3. Applying our proposed method on other datasets

In this section, we present the results of applying our proposed method on two famous heart disease datasets from Data Mining Repository of University of California, Irvine (UCI) [67]: "Cleveland" and "Hungarian". We want to compare the performance of our proposed algorithm with some other methods.

Cleveland dataset was collected by Robert Detrano at V.A. Medical center. Machine learning researchers often use the Cleveland database. Class distributions are 54% healthy and 46% CAD [68].

Hungarian dataset was collected by Andras Janosi at the Hungarian Institute of Cardiology, Budapest. Class distributions are 62.5% healthy and 37.5% CAD [69].

In those datasets, totally, 14 attributes such as age, sex, chest pain type, resting blood pressure, serum cholesterol in mg/dl, fasting blood sugar, resting electrocardiographic results, maximum heart rate achieved, exercise induced angina, ST depression, slope of the peak exercise ST segment, number of major vessels colored by fluoroscopy, thal and diagnosis of heart disease are presented.

One of the weaknesses of Cleveland and Hungarian heart disease datasets is that they did not determine stenosis of LAD, LCX and RCA features. We collected our new dataset (extended Z-Alizadeh sani) to cover this weakness.

To show our proposed method does not depend on special dataset, we must test our methods on other datasets. Unfortunately, other datasets do not have LAD, LCX and RCA features. So, we selected common features of UCI heart disease and Z-Alizadeh Sani datasets. Then, we trained our proposed method on these features of extended Z-Alizadeh Sani dataset and test it on these features of UCI heart disease datasets. The results are illustrated in Tables 6 and 7. As it is clear from the tables, our proposed method could increase the performance. This shows that if LAD, LCX and RCA features were collected in UCI heart disease datasets, it could

Table 6
Experimental results for the proposed method and other methods on the Cleveland dataset.

Method	Accuracy (%)	Sensitivity (%)	Specificity (%)
Rough-Fuzzy Classifier [70]	46.48	69	27
C4.5 [71]	78.54	N.R.[a]	N.R.
FDT [71]	77.55	N.R.	N.R.
Neural Network	85.52	59	94
RBF [35]	60.00	N.R.	N.R.
FOIL [35]	64.00	N.R.	N.R.
MLP + BP [35]	65.60	N.R.	N.R.
Naïve Bayes [35]	83.60	N.R.	N.R.
Neural networks [35]	89.01	81	96
C4.5 [72]	81.11	N.R.	N.R.
Naïve Bayes [72]	81.48	N.R.	N.R.
BNND [72]	81.11	N.R.	N.R.
BNNF [72]	80.96	N.R.	N.R.
AIRS [73]	84.50	N.R.	N.R.
Weighted fuzzy rules [74]	86.35	76	49
Our Proposed Method	**93.06**	94	92

[a] N.R. means "not reported".

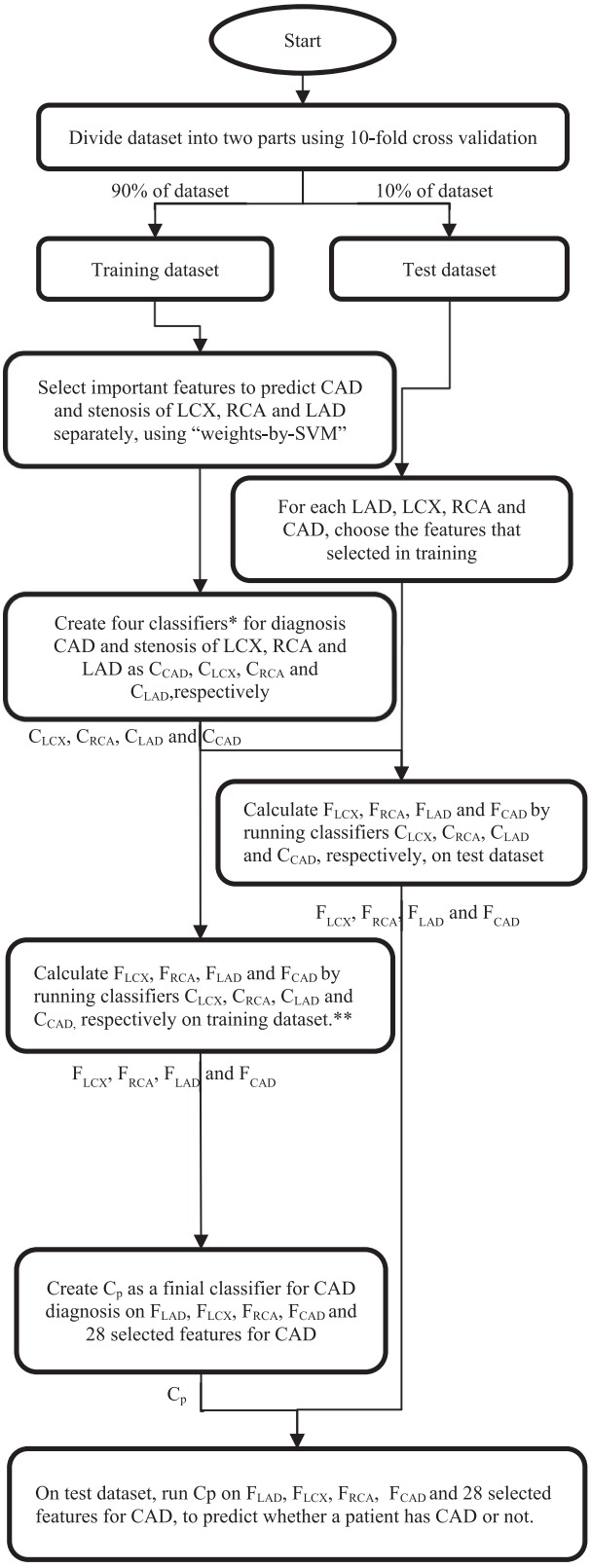


Fig. 1. The Proposed feature engineering method. * The classifier can be any classification algorithm. ** F_x is probability that x is stenosed.

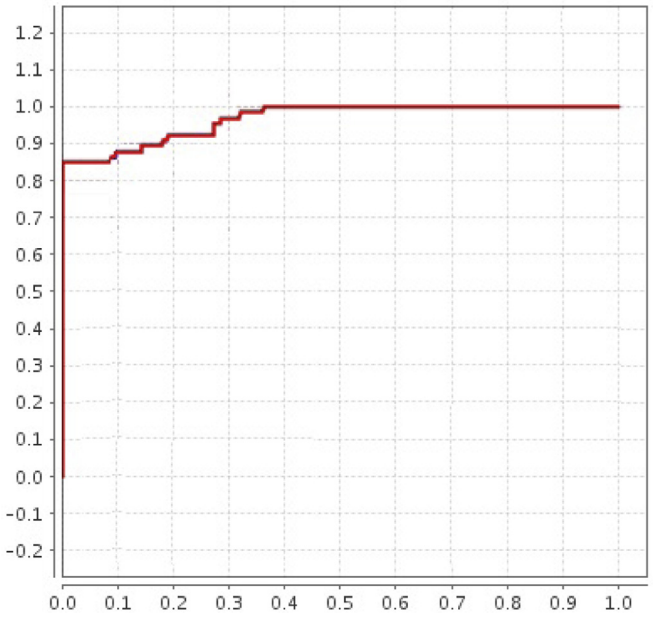


Fig. 2. ROC curve of our proposed method.

Table 7
Experimental results for the proposed method and other methods on the Hungarian dataset.

Method	Accuracy (%)	Sensitivity	Specificity
Rough-fuzzy classifier [70]	42.42	0.64	0.29
C4.5 [71]	78.57	N.R.	N.R.
FDT [71]	78.23	N.R.	N.R.
Neural network	84.50	0.70	0.93
Weighted fuzzy rules [74]	49.93	0.40	0.74
Naïve Bayes	78.23	0.85	0.74
MLP	81.63	0.90	0.77
RBF	83.33	0.90	0.80
RFC	83.20	0.85	0.77
Our proposed method	**88.77**	0.88	0.89

improve the performance of CAD detection. It is one of the weaknesses of these datasets. The smaller number of original features which were used in these two datasets caused lower performance with respect to extended Z-Alizadeh sani dataset. Meanwhile, better features collected in extended Z-Alizadehsani dataset improved CAD diagnosis accuracy.

5. Discussion

In this research, we achieved high sensitivity and accuracy rates for diagnosis of CAD. In our previous research, there were 303 record of patients [60]. However, in the new one, it is extended to 500 records. Our main goal for increasing the number of records from 303 to 500 was achieving to more confident results. By increasing the number of records, we got almost the same results. It shows that the robustness of our algorithm and the results achieved on 303 records were not stochastic.

This method has less costs and side effects compared to angiography for patients. In our proposed method, we used LAD, LCX and RCA stenosis prediction for CAD diagnosis for the first time in the literature. We did it because a patient has CAD if and only if at least one of these arteries is stenotic. The result showed that using LAD, LCX and RCA stenosis prediction can improve accuracy significantly.

We reached the sensitivity value of 100% and accuracy rate of 96.40%. To the best of our knowledge, these are the highest sensi-

tivity and accuracy rates in the literature. The rate of 100% sensitivity strongly suggest that negative result of our proposed method can be used as a very strong counter-indicator for angiography.

In addition, the sensitivity rate of 100% makes the method highly applicable. In fact, according to the results from the extended Z-Alizadeh Sani dataset, a patient can be tested with this method first. In case of negative result of the method, he/she does not need to undergo angiography. Therefore, the side effects and costs of angiography can be avoided. Otherwise, angiography is recommended for him/her to determine the exact location and percent of the stenosis.

6. Conclusion and future work

The present study proposes a novel machine learning algorithm for CAD diagnosis. For this purpose, three classifiers were used to predict the stenosis of coronary arteries, i.e., LAD, LCX and RCA to get higher accuracy for CAD diagnosis. Our proposed method achieved excellent accuracy and sensitivity for CAD diagnosis. For more reliable results, we also increased the number of samples from 303 to 500 cases. With respect to high cost and side effects of angiography, removing the need to undergo angiography for most of healthy patients is an invaluable work in medicine.

According to physicians, a person has CAD if at least one of LAD, LCX, and RCA coronary arteries has stenosis more that 50%. However, if stenosis is between 30% and 50%, the patient has minimal CAD. Checking the false predictions of the algorithm, i.e., the healthy individuals determined as patients, showed that all of them had minimal CAD. So the error of this algorithm is mostly because of predicting some healthy people who have minimal CAD as CAD patients. Although, our proposed method has excellent accuracy and sensitivity, its specificity is not competitive with angiography. Our proposed method can distinguish all CAD patients but not all non-CAD patients.

The prediction of the severity of the stenosis is the natural extension of this work. Extending the dataset to achieve more reliable result and using some other new methods such as deep learning are another work that must be done. We will also try to achieve the specificity rate of angiography.

Conflict of interest

The authors declare that there is no conflict of interest related to this paper.

Appendix A

We extracted some important rules for LAD, LCX and RCA stenosis using association rule mining. These rules that indicate the relationship between the stenosis of the arteries have been reported in Appendix A.

Rules produced by the association rule mining

We have produced rules by the association rule mining techniques to give more intuition about the dataset. These rules indicate the relationship between the stenosis of the arteries. Also the relationship between the stenosis of the arteries and the sex of the patients are discussed. In these rules, C represents the Confidence.

Relation between arteries:

1.	[LAD ≥ 50%, LCX ≥ 50 %] = > [RCA ≥ 50%],	C = 0.65625
2.	[LAD ≥ 50%, RCA ≥ 50 %] = > [LCX ≥ 50%],	C = 0.72414
3.	[RCA ≥ 50%, LCX ≥ 50 %] = > [LAD ≥ 50%],	C = 0.86301
4.	[RCA ≥ 50 %] = > [LCX ≥ 50%],	C = 0.64035
5.	[RCA ≥ 50 %] = > [LAD ≥ 50%],	C = 0.76316
6.	[LAD ≥ 50 %] = > [RCA ≥ 50%],	C = 0.49153
7.	[LAD ≥ 50 %] = > [LCX ≥ 50%],	C = 0.54237
8.	[LCX ≥ 50 %] = > [RCA ≥ 50%],	C = 0.61345
9.	[LCX ≥ 50 %] = > [LAD ≥ 50%],	C = 0.80672
10.	[LCX ≥ 50 %] = > [LAD ≥ 50%, RCA ≥ 50%],	C = 0.52941
11.	[LAD ≥ 50 %] = > [LCX ≥ 50%, RCA ≥ 50%],	C = 0.35593
12.	[RCA ≥ 50 %] = > [LAD ≥ 50%, LCX ≥ 50%],	C = 0.55263

Rule 2 shows that the probability of LCX stenosis is high given that LAD and RCA are stenotic. Similarly, rule 3 shows that the probability of LAD stenosis is high given that LCX and RCA are stenotic. Moreover, rule 9 states that LAD stenosis probability is high given that the LCX is stenotic.

Besides the extracted rules, the following results can be seen in the extended Z-Alizadeh Sani dataset which has 500 samples: RCA artery of 36.4% patients, LCX artery of 39.2% of patients and LAD artery of 58.4% of patients is stenotic. The both LAD and LCX arteries of 31.6% of patients are stenotic. This shows that most of whom their LAD artery is stenotic, their LCX artery is stenotic too. Finally, LAD and RCA arteries of 28.7% of patients are stenotic together, RCA and LCX arteries of 24.0% of patients are stenotic together and 20.7% of patients have the stenosis of all the three arteries.

Male and female rules:

1.	[Sex = Female] = > [LAD ≥ 50%],	C = 0.52755
2.	[Sex = Female] = > [LCX ≥ 50%],	C = 0.3307
3.	[Sex = Female] = > [RCA ≥ 50%],	C = 0.30708
4.	[Sex = Male] = > [LAD ≥ 50%],	C = 0.625
5.	[Sex = Male] = > [LCX ≥ 50%],	C = 0.4375
6.	[Sex = Male] = > [RCA ≥ 50%],	C = 0.42613

These rules show that probability of stenosis is lower for women in comparison with men. Also in both women and men, LCX and RCA stenosis is less probable than LAD stenosis. Moreover, the probability of RCA stenosis is less than LCX, and the LAD stenosis is almost 1.6 of the two other arteries.

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